Elacestrant real-world progression-free survival (rwPFS) of adult patients with ER+/HER2-, advanced breast cancer: a retrospective analysis using insurance claims in the United States

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BACKGROUND

- Endocrine therapy (ET) + CDK4/6i is the standard-of-care (SOC) for 1st-line treatment of ER+/HER2- advanced or metastatic breast cancer (mBC).[1-3]
- In current practice, sequential endocrine monotherapy or combination therapies are used in the 2nd-line post ET + CDK4/6i as recommended by guidelines.[2,4,5] Combination regimens in ≥2nd-line mBC can be associated with significant toxicities and high discontinuation rates.[6-12]
- *ESR1* mutations represent a type of acquired resistance that eventually emerges in up to 50% of patients after 1st-line treatment for ER+/HER2- mBC.[13-20] Current guidelines recommend testing using liquid biopsy for the emergence of *ESR1* mutations at each disease progression.[21]
- **In the EMERALD trial, patients with *ESR1*-mutated tumors receiving single-agent elacestrant had a 45% reduction in risk of progression or death vs SOC ET; median progression-free survival (mPFS) was 3.8 months with elacestrant vs 1.9 months with SOC ET (HR=0.55; 95% CI: 0.39-0.77; *P*=0.0005).[22]**
  - **In patients with ≥12 months of prior ET + CDK4/6i, mPFS with elacestrant was 8.6 months vs 1.9 months with SOC (HR=0.41; 95% CI: 0.26-0.63)[23]**
- Based on the EMERALD trial, elacestrant was approved by the FDA in January 2023 for the treatment of postmenopausal ER+/HER2- *ESR1*-mutated advanced or metastatic breast cancer with disease progression following at least one line of ET.
- Sufficient time has passed since approval to characterize the real-world use of elacestrant in the current treatment landscape.
- **This study aims to describe the real-world PFS (rwPFS) of elacestrant overall, as well as in patients with no prior exposure to fulvestrant, only 1-2 lines of ET, ≥3 lines of ET, and/or visceral metastasis.**

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METHODS

Database	Komodo Research Database (KRD+)
Study population	▪ Adults with diagnosed ER+/HER2- mBC who initiated elacestrant in a real-world setting were identified between January 2023 and May 2024 ▪ Elacestrant is indicated for the treatment of postmenopausal women and men with ER+/HER2- *ESR1*-mutated advanced or mBC with disease progression following at least one line of ET. *ESR1*-mutation status was not available in the KRD+ at the time of analysis, but it was assumed patients received treatments according to their approved use
Index date	Time of first administration of elacestrant
Demographic information at index date	▪ Metastatic sites identified during the 6-month baseline period prior to the index date ▪ Prior treatments received in mBC setting
Definition of rwPFS	Time from index date until earliest outcome (start of next line of therapy for mBC to disease progression or death) assessed using time-to-event and Kaplan-Meier methods
Censoring rule	Patients were censored at first occurrence of: ▪ End of data availability, OR ▪ End of health plan enrollment
Limitations	▪ Real-world differences with routine clinical care versus prospective trials may be due to multiple factors impacting patient selection: treatment heterogeneity, tumor burden, genomic profile, and follow-up duration from the time of first elacestrant treatment. Variations in patient populations and baseline characteristics might require further research.

ET, endocrine therapy; KRD+, Komodo Research Database; mBC, metastatic breast cancer; rwPFS, real-world progression-free survival.

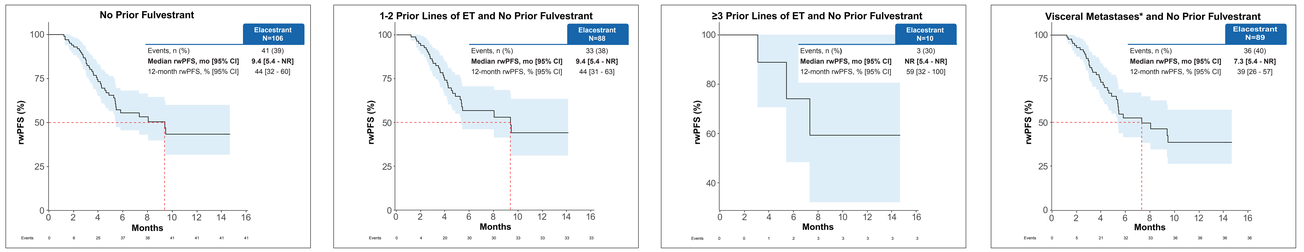
RESULTS

Overall demographics and clinical characteristics

Parameters	N=276
Median age (years) as of elacestrant initiation	62.8
Female, n (%)	276 (100)
Race/Ethnicity, n (%)	
White (non-Hispanic)	148 (54)
Black or African American	26 (9)
Hispanic or Latino	20 (7)
Unknown	17 (6)
Asian or Pacific Islander	11 (4)
Other	4 (1)
Not characterized	50 (18)
Visceral metastasis (sites other than bone/lymph node only), n (%)	237 (86)
Number of prior lines of ET, n (%)	
0*	8 (3)
1	71 (26)
2	95 (34)
3	57 (21)
4	26 (9)
5	8 (3)
6	6 (2)
≥7	5 (2)
Prior mBC treatment, n (%)	
ET	268 (97)
CDK4/6i	248 (90)
ET ± CDK4/6i ≥12 months from first mBC diagnosis to index date	242 (88)
Fulvestrant	170 (61)
Chemotherapy	92 (33)

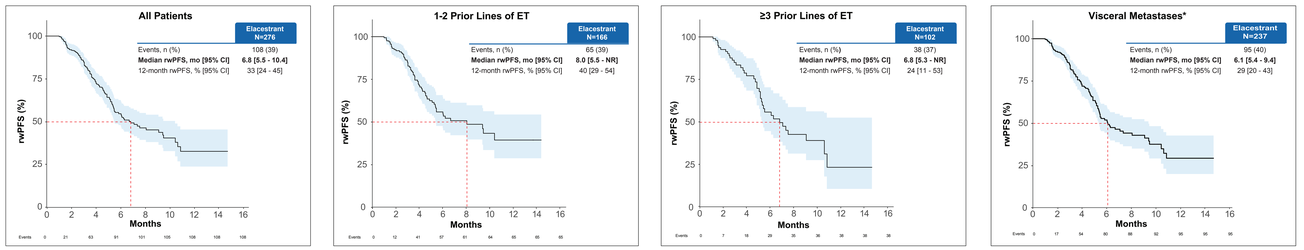
*Patients with 0 prior lines of ET from mBC diagnosis to index all received ET in the pre-metastatic setting.
Abbreviations: CDK4/6i, cyclin-dependent kinase 4 and 6 inhibitors; ET, endocrine therapy; mBC, metastatic breast cancer.

No prior fulvestrant: elacestrant showed prolonged rwPFS in a real-world setting



*Visceral mets (sites other than bone/lymph node only). Abbreviations: CI, Confidence Interval; ET, endocrine therapy; NR, not reached; rwPFS, real-world Progression-free survival.

Overall population: elacestrant showed prolonged rwPFS across all patients in a real-world setting



*Visceral mets (sites other than bone/lymph node only). Abbreviations: CI, Confidence Interval; ET, endocrine therapy; NR, not reached; rwPFS, real-world Progression-free survival.

Elacestrant showed a consistent rwPFS benefit among all patients and across clinically relevant subgroups

	N (%)	Events, n (%)	Median rwPFS, months [95% CI]	12-month rwPFS, % [95% CI]
No prior fulvestrant *(38% of overall population)*	106 (100)	41 (39)	**9.4** [5.4 - NR]	44 [32 - 60]
1-2 prior lines of ET	88 (83)	33 (38)	**9.4** [5.4 - NR]	44 [31 - 63]
≥3 prior lines of ET	10 (9)	3 (30)	**NR** [5.4 - NR]	59 [32 - 100]
Visceral metastases (all lines)*	89 (84)	36 (40)	**7.3** [5.4 - NR]	39 [26 - 57]
Overall population *(includes prior fulvestrant)*	276 (100)	108 (39)	**6.8** [5.5 - 10.4]	33 [24 - 45]
1-2 prior lines of ET	166 (60)	65 (39)	**8.0** [5.5 - NR]	40 [29 - 54]
≥3 prior lines of ET	102 (37)	38 (37)	**6.8** [5.3 - NR]	24 [11 - 53]
Visceral metastases (all lines)*	237 (86)	95 (40)	**6.1** [5.4 - 9.4]	29 [20 - 43]

*Visceral mets (sites other than bone/lymph node only), all treatment lines.
Abbreviations: CI, Confidence Interval; ET, endocrine therapy; NR, not reached; rwPFS, real-world Progression-free survival.

CONCLUSIONS

- In a real-world setting, elacestrant showed a consistent rwPFS benefit among all ER+/HER2- mBC patients and across clinically relevant subgroups:

mPFS (months)	Patients with no prior fulvestrant (n=106)	All patients (n=276)
Overall mPFS	**9.4**	**6.8**
1-2 lines of ET	**9.4**	**8.0**
Visceral metastasis	**7.3**	**6.1**

- **The rwPFS with elacestrant exceeded the overall mPFS reported in the EMERALD study.**
- **The rwPFS with elacestrant is longer when patients are treated in earlier lines.**
- Future analysis of additional factors such as coexisting mutations (ie, PI3K/AKT1/PTEN alterations) and other clinically relevant subgroups will provide meaningful insights to help inform real-world clinical decision-making.